Automated Classroom Attendance: Foundations, Metric Learning & Engineering Problem-Solving

A Comprehensive Pedagogical Guide to Hyperspherical ArcFace Embeddings, Anchor-Free CNNs, Multi-Scale IoM Deduplication, and Mobile Edge Optimization

Executive Summary & Pedagogical Scope: *This technical document provides both a rigorous mathematical exposition and an empirical engineering chronicle of the AURA platform. It covers: (1) discrete roll call time complexity modeling vs. open-set facial verification, (2) OpenCV YuNet anchor-free spatial detection and GIoU loss, (3) SVD-based affine Procrustes alignment, (4) InsightFace ArcFace additive angular margin on the 512D hypersphere, (5) high-throughput cosine matrix verification, and (6) in-depth case studies of real-world edge challenges (including the 'Krishna Agarwal' hubness false positive cascade, Android 48MP resolution explosion, and multi-scale anchor dual boxes).*

1. Problem Modeling: Discrete Roll Calls vs. Continuous Biometric Verification

Traditional classroom attendance relies on discrete human sampling—either vocal roll-calling or paper sign-in rosters. Formally, consider a course enrollment set $S = \{s_1, s_2, ..., s_K\}$ of K students. Manual roll calling represents a linear search $O(K)$ executed at an average human latency of $t_{call} \approx 8$ s per student. For a classroom of $K = 100$, total instructional time expenditure reaches:

$$T_{manual} = \sum_{i=1}^{K} t_{call}(s_i) \approx 100 \times 8 \text{ s} = 800 \text{ s} \approx 13.3 \text{ minutes}$$

Over a standard university semester of $N_{sessions} = 40$ lectures, cumulative lost instructional time exceeds **8.8 hours**. Furthermore, paper roll sheets provide zero cryptographic authentication, exhibiting an empirical proxy vulnerability rate exceeding 15%. AURA models attendance as an optimal open-set facial verification problem over an optical image frame $I \in \blacksquare^{H \times W \times 3}$.

$$f: I \rightarrow \blacksquare = \{(b_k, y_k, s_k)\}_{k=1}^{M}, \; y_k \in S \cup \{\text{Unknown}\}$$

where M is the number of detected faces, $b_k = (x, y, w, h)$ is the bounding box coordinates, y_k is the identified enrolled student (or Unknown), and $s_k \in [0, 1]$ is the cosine confidence score.

2. OpenCV YuNet: Anchor-Free CNN Multi-Scale Face Detection

YuNet is an ultra-lightweight, anchor-free convolutional neural network designed specifically for real-time edge face detection. Unlike legacy anchor-based detectors (e.g., RetinaFace, Faster R-CNN) that evaluate dense geometric anchor priors across multiple scales (incurring high computational overhead), YuNet operates directly on feature map grid points.

2.1 Anchor-Free Spatial Regression & Generalized IoU Loss

For each feature pyramid grid location (x_g, y_g), YuNet predicts a 4-dimensional bounding box distance tuple (l, t, r, b) representing offsets to the left, top, right, and bottom boundaries. To ensure scale-invariant gradient descent, bounding box regression is optimized using Generalized Intersection over Union (GIoU) loss:

$$\blacksquare_{GIoU} = 1 - IoU(B, B\blacksquare) + |C \setminus (B \cup B\blacksquare)| / |C|$$

where B is the ground truth bounding box, $B\blacksquare$ is the predicted bounding box, and C is the smallest convex hull enclosing both B and $B\blacksquare$. GIoU solves the vanishing gradient limitation of standard IoU when predicted and target boxes do not

overlap, providing continuous smooth gradients.

2.2 5-Point Facial Landmark Regression

Simultaneously, YuNet regresses 5 primary facial landmarks: left eye p_1, right eye p_2, nose tip p_3, left mouth corner p_4, and right mouth corner p_5 using smooth L1 loss:

$$\blacksquare_{lmk} = \Sigma_{i=1}^{5} \text{smooth}_{L1}(p_i - p\blacksquare_i)$$

These 5 landmarks are critical for decoupling head pose, camera tilt, and in-plane perspective distortions prior to identity verification.

3. Geometric Affine Normalization & SVD Procrustes Alignment

Raw face crops exhibit substantial pitch, yaw, roll, and distance variations. Feeding unaligned face patches directly into a deep metric recognizer drastically degrades inter-class discriminability. AURA normalizes all detected faces onto a standardized canonical 112 × 112 coordinate frame using an optimal affine transformation.

Let $\mathbf{P} = \{p_i\}_{i=1}^{5}$ be the observed landmarks and $\mathbf{Q} = \{q_i\}_{i=1}^{5}$ be the canonical frontal facial template coordinates. We seek an affine similarity transformation matrix $T \in \blacksquare^{2 \times 3}$ minimizing the Procrustes Euclidean landmark residual:

$$\min_T \Sigma_{i=1}^{5} || T \cdot [p_i, 1]^T - q_i ||^2$$

Using Singular Value Decomposition (SVD) of the centered covariance matrix $\Sigma_{PQ} = P^T Q = U D V^T$, the optimal rotation is solved as $R = U V^T$. Once T is computed, bilinear spatial interpolation maps the pixel grid into the canonical frontal patch $I_{aligned} \in \blacksquare^{112 \times 112 \times 3}$.

4. InsightFace ArcFace: Additive Angular Margin Loss on a Hypersphere

The machine learning backbone of AURA is the ArcFace (ResNet-50) deep convolutional neural network. ArcFace reformulates feature learning entirely on the surface of a unit hypersphere S^{d-1} in $\blacksquare^d$ (where $d = 512$). Both the class weight vectors W_j and feature embeddings x_i are $\blacksquare_2$-normalized:

$$||W_j||_2 = 1, \; ||x_i||_2 = 1 \Rightarrow W_j^T x_i = \cos(\theta_j)$$

where θ_j is the geometric angle between feature vector x_i and class centroid W_j on the hypersphere. To enforce an explicit angular decision boundary, ArcFace injects an additive angular margin penalty m directly into the target angle θ_{y_i}:

$$\blacksquare_{ArcFace} = - (1/N) \Sigma_{i=1}^{N} \log \left[e^{s \cdot \cos(\theta_{y_i} + m)} / \left(e^{s \cdot \cos(\theta_{y_i} + m)} + \Sigma_{j \neq y_i} e^{s \cdot \cos(\theta_j)} \right) \right]$$

where **s = 64** is the feature scaling radius and **m = 0.5 rad (≈ 28.6°)** is the additive angular margin penalty, guaranteeing strict intra-class geodesic concentration and maximizing inter-class separation.

Loss Function	Metric Formulation	Decision Boundary	Intra-Class Geometry
Softmax	W^T x = ||W||||x|| cos θ	W_1^T x = W_2^T x	High variance / scattered
SphereFace	cos(m·θ)	cos(m·θ_1) = cos(θ_2)	Multiplicative margin (unstable)
CosFace	cos(θ) - m	cos(θ_1) - m = cos(θ_2)	Cosine margin (linear)
ArcFace (AURA)	cos(θ + m)	cos(θ_1 + m) = cos(θ_2)	Exact geodesic angular margin

5. Vector Verification & Cosine Matrix Matching

During live inference, ArcFace maps any normalized aligned face crop $I_{aligned}$ into a unit vector embedding $e \in \blacksquare^{512}$, such that $||e||_2 = 1$. The degree of biometric similarity between a probe face e_{probe} and an enrolled student vector $e_{gallery}$ simplifies to their vector dot product:

$$\mathbf{sim(e_{probe}, e_{gallery}) = e_{probe} \cdot e_{gallery} = e_{probe}^{T} e_{gallery}}$$

For a classroom course with K enrolled embedding vectors, all registered embeddings are stacked into a matrix $E_{gallery} \in \blacksquare^{K \times 512}$. When an optical frame containing M detected students is processed, the probe matrix $P \in \blacksquare^{M \times 512}$ computes all pairwise similarities simultaneously via a single BLAS GEMM matrix multiplication:

$$\mathbf{S = P \cdot E_{gallery}^{T} \in \blacksquare^{M \times K}}$$

On modern Intel/AMD CPU hardware with AVX-512 extensions, computing S for $M = 100$ faces and $K = 500$ enrolled vectors resolves in less than **3.2 ms**.

6. Case Study 1: The 'Krishna Agarwal' False Positive Cascade (Hubness Failure)

During mobile phone attendance testing, an unexpected phenomenon occurred: multiple unrelated students and strangers were repeatedly misclassified as enrolled student **Krishna Agarwal** with match confidence exceeding 60%. Systematic debugging revealed a classic failure mode in deep metric learning known as the **Hubness Problem**.

6.1 Root Cause: The Area-Only Registration Trap

When Krishna was initially enrolled from a phone photo, YuNet localized two bounding boxes:
• **Face 0 (Real Face)**: Confidence = 0.914, Bounding Box = [140, 471, 219, 312], Area = 68,328 px^2.
• **Face 1 (Shirt Collar / Neck)**: Confidence = 0.594, Bounding Box = [248, 708, 320, 376], Area = 120,320 px^2.
Because the enrollment code previously selected the primary face purely by bounding box area $w \times h$, it selected the non-face collar patch!

6.2 Hyperspherical Collapse to a Central Hub Vector

When a non-face image patch is passed into an ArcFace network trained exclusively on human faces, the lack of distinct facial landmarks causes the intermediate feature maps to collapse toward the global feature centroid on S^{511}. In high-dimensional metric space, such a collapsed vector acts as a **'Hub Vector'**: it possesses high cosine similarity (0.45 to 0.64) with a broad distribution of unrelated face embeddings. On a 114-student classroom benchmark photo, this single corrupted embedding matched **16 random strangers** above the threshold!

6.3 The Mathematical Solution: Multi-Factor Face Quality & Ensemble Scoring

Two structural defenses were engineered to permanently eradicate hubness vulnerabilities:
1. **Multi-Factor Face Quality Metric**: Candidate selection now combines confidence, centrality, and scale:

$$\mathbf{Quality(f) = (conf)^2 \times [1 - dist(center_{face}, center_{image})] \times \sqrt{(Area / Area_{image})}}$$

2. **Ensemble Student-Level Pooling**: For students with multiple reference photos, similarity is no longer determined by a vulnerable single maximum. Instead, the engine computes a blended ensemble of the top-2 similarities:

$$\mathbf{Score_{ensemble} = 0.75 \cdot s_{top1} + 0.25 \cdot s_{top2}}$$

If a single embedding produces an outlier match while the student's other photos show zero correlation ($s_{top2} < 0.15$), the match is penalized and rejected. Following database re-extraction, false positive matches on the 114-face classroom photo dropped from 16 to exactly **0**.

7. Case Study 2: Android 48MP Resolution Explosion & Web Security

While laptop webcam attendance functioned smoothly, testing from mobile devices revealed two platform-specific disparities: (1) high mobile camera resolution causing neural network saturation, and (2) browser camera permission blocks.

7.1 Receptive Field Saturation at 4000 × 3000 Resolution

Modern smartphones capture native photos at 12MP to 48MP (e.g. 4000 × 3000). When fed into YuNet without preprocessing, the convolutional receptive fields saturated. Benchmark tests revealed that a 4000 × 3000 input caused YuNet to detect **11 phantom faces** on a single individual, with inference latency surging from 73 ms to **1131 ms**. Rescaling the image to a standardized maximum dimension of 1600 px restored clean detection (1 face) and dropped latency to **138 ms**.

7.2 Client-Side Canvas Pre-Compression

Uploading raw 15 MB mobile photos exhausted mobile browser RAM and created 5–8 second Wi-Fi transfer delays. A client-side HTML5 canvas optimizer (*optimizeImageForUpload*) was integrated into the frontend. Photos are downscaled to 1600 px client-side before transmission, cutting network payload to ~350 KB, normalizing EXIF orientation, and reducing transfer time to **< 100 ms**.

7.3 Chromium W3C Insecure Origin Policy

On Android Chrome, accessing the server over LAN HTTP (e.g. *http://192.168.x.x:8000*) strictly disables *navigator.mediaDevices.getUserMedia*. The web interface was enhanced with defensive detection: on mobile HTTP, the UI automatically routes users to **Group Photo AI Upload**, where native camera inputs (*capture='environment'*) operate unrestricted with 1-tap capture speed.

8. Case Study 3: Dual Bounding Boxes on Close-Up Faces & IoM Deduplication

When students took close-up selfie portraits occupying >50% of the camera frame, the UI rendered **two overlapping green boxes** on the same face (e.g., Aman 48% on upper face, Aman 47% on lower face).

8.1 Feature Pyramid Anchor Grid Collision

On large faces, adjacent Feature Pyramid Network (FPN) levels in YuNet simultaneously detected overlapping sub-regions: Anchor level P_4 localized the forehead and eyes, while level P_5 localized the eyes, nose, and mouth. Because the vertical shift between the two boxes reduced their standard IoU to ~18%, standard Non-Maximum Suppression (NMS threshold = 0.30) treated them as two separate people.

8.2 Intersection over Minimum (IoM) Suppression Formulation

To resolve multi-scale anchor collisions, AURA introduced an **Intersection over Minimum (IoM)** suppression filter:

$$\text{IoM}(B_1, B_2) = \text{Area}(B_1 \cap B_2) / \min(\text{Area}(B_1), \text{Area}(B_2))$$

If $IoM(B_1, B_2) > 0.30$, the detector recognizes that the smaller box is substantially enclosed within the larger face region, suppressing the duplicate. In addition, an identity constraint enforces that a single enrolled student cannot possess two overlapping detections, guaranteeing exactly **one clean box** per student.

9. Complete Engineering Problem-Solving Chronology

The following matrix summarizes the key engineering hurdles encountered throughout the project lifecycle, their theoretical root causes, and the deployed production fixes:

Hurdle	Manifestation	Root Cause	Engineering Solution

1. Windows Locks	Errno 13 Permission Denied on PPTX	PowerPoint exclusively locks open files	Dynamic timestamped file generation & FastAPI streamin
2. Console Crash	UnicodeEncodeError on cp1252 terminal	Windows console fails on special characters like ✓ or ✗	Clean ASCII logging tags ([MATCH], [UNKNOWN])
3. Portrait Drop	Divyansh registration rejected (400)	Phone EXIF rotation tilt & multi-face background confusion	ImageOps.exif_transpose + 360° fallback rotation search
4. Hubness Storm	Strangers classified as Krishna Agarwal	Area-only sort picked collar patch; ArcFace hubness curse	Multi-factor quality metric + Top-2 ensemble pooling
5. Mobile Bloat	Phone UI lag; 11 phantom detections	4000x3000 native phone resolution saturated CPU & caused OOM	Client-side canvas compression + backend 1600px guard
6. Dual Box Bug	Two overlapping boxes on close-up face	FPN multi-scale anchors fired; vertical shift evaded IoU	Intersection over Minimum (IoM > 0.30) deduplication

10. Pedagogical Takeaways & Production Guidelines

• **Metric Learning Quality Dictates Accuracy:** A deep embedding space is only as good as the crops registered into it. Strict quality scoring on registration photos is essential to prevent hubness collapse.

• **Anchor Scales Must Match Receptive Fields:** Feeding ultra-high-resolution images (e.g. 48MP) into standard face detectors degrades both speed and accuracy; input normalization is mandatory.

• **IoM Trumps IoU on Large Objects:** Standard IoU fails when detectors output shifted sub-components of large objects. Intersection over Minimum (IoM) is the mathematically correct metric for multi-scale containment.

• **Edge Defense-in-Depth:** Combining client-side image compression, server-side dimension guards, and ensemble vector verification yields sub-second classroom attendance that is robust across laptops and mobile phones.